

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Data Interpretation

Weightage: 20%

QUESTIONS:

Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
  <head>
    <title>Test Page</title>
  </head>
  <body>
    <div>
```

```
<p>Hello World
</div>
</body>
</html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph

</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned

6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

The screenshot displays a web browser window with the address bar showing the file path: `C:/Users/tejas/OneDrive/Desktop/ip/Assessment%20tool2/index.html`. The page content is divided into several sections:

- Possible delays**: Delays can occur during DNS resolution, connection and TLS setup, redirects, server queues, database/API calls, response transfer, and downloading or rendering CSS, JavaScript, fonts, images, and third-party resources.
- Role of DNS**: DNS is the naming layer that converts a human-readable host name into the network address needed to reach the server. If DNS is slow or unavailable, the website cannot normally be reached.
- Performance optimizations**: Use reliable DNS and caching, HTTPS with HTTP/2 or HTTP/3, keep-alive connections, a CDN, compression (Brotli/Gzip), cache-control headers, optimized and lazy-loaded images, fewer redirects, deferred non-critical scripts, and efficient server/database operations.
- Declaration**: A form for the student to certify that the submission is their original work and that they have adhered to the guidelines. It includes fields for the student's name/registration number and a signature line.

Below the declaration section, the page header changes to a dark blue background with the text "SIMATS ENGINEERING" and "CO1-AT2 - Data Interpretation". The sub-header "Internet Programming" is also present. The main content area features a form for "Student Name" and "Register Number". Below this, a section titled "1. HTTP Response Analysis" is shown, which includes a "Given response" block and a discussion on "Incorrect or invalid fields" and "Content-Length issue and impact".

CSA43 Assessment Tool 2 - Dail

C:/Users/tejas/OneDrive/Desktop/ip/Assessment%20tool2/index.html

Content-Length must be a non-negative decimal number giving the exact number of bytes in the message body. A negative value is invalid. The browser cannot reliably identify the end of the body, so it may reject the response or be unable to reuse the keep-alive connection safely.

Status-code correctness

200 OK is the correct status for a successfully processed request that returns an HTML representation. However, the malformed header makes the response itself invalid.

Corrected response

```
HTTP/1.1 200 OK
Date: Wed, 01 Apr 2026 18:00:00 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 150
Connection: keep-alive
```

The date has also been corrected: 1 April 2026 is a Wednesday. The length must equal the real response-body size; otherwise use a valid transfer-framing method such as chunked encoding.

2. DOM Structure Analysis

```
<html>
<head><title>Test Page</title></head>
<body><div><p>Hello World</div></body>
</html>
```

2. DOM Structure Analysis

```
<html>
<head><title>Test Page</title></head>
<body><div><p>Hello World</div></body>
</html>
```

Missing or improperly closed element

The opening <p> tag has no closing </p> tag.

Browser interpretation and rendering impact

HTML parsers recover from this error by implicitly closing the paragraph before the closing </div>. It will normally render as a paragraph inside the div. In larger documents, implicit closing can create unexpected DOM structures and affect CSS, JavaScript, spacing, and accessibility.

Valid corrected DOM

```
<html>
<head><title>Test Page</title></head>
<body>
  <div>
    <p>Hello World</p>
  </div>
</body>
</html>
```

CSA43 Assessment Tool 2 - Dashboard

C:/Users/tejas/OneDrive/Desktop/ip/Assessment%20tool2/index.html

3. GET vs POST in a Login System

Aspect	GET	POST
Transmission	Appends parameters to the URL query string.	Places form data in the HTTP request body.
Visibility	Can appear in history, logs, bookmarks, referrer headers, and shared links.	Not displayed in the URL, but still needs HTTPS for protection in transit.
Login suitability	Not suitable.	Appropriate for submitting credentials.

Analysis

POST is more suitable because it does not expose credentials in the URL and login establishes authentication state. "In the request body" is not the same as encrypted; HTTPS is mandatory.

Security risks of GET

A password in the query string can be retained by browser history, server and proxy logs, analytics tools, bookmarks, screenshots, and `referrer` headers. This can expose credentials to unauthorized people or systems.

Secure login practices

- Use POST over HTTPS; never send credentials in a URL.
- Hash passwords with Argon2id or bcrypt and a unique salt; never store plain-text passwords.
- Use Secure, HttpOnly, SameSite cookies and regenerate the session ID after login.
- Validate input server-side, rate-limit failed attempts, use generic error messages, and never log passwords.

4. HTML Code Inspection

Errors identified

The `<h1>` and `<p>` tags have no explicit closing tags. The `<img>` element is a void element, so it does not require a closing tag, but it lacks alternative text. Providing `alt` text is important for accessibility.

Browser handling

The browser will implicitly close the heading when it encounters the paragraph, then implicitly close the paragraph at the end of the body. It may render, but the invalid markup can cause inconsistent styling and DOM behaviour.

Corrected code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Sample</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is a paragraph.</p>
  
</body>
</html>
```

Code of Conduct:

I C. Tejashwi (192372043) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand